Corpus Statistics Report

Hybrid Retrieval System (Phase 1) - arXiv cs.AI corpus

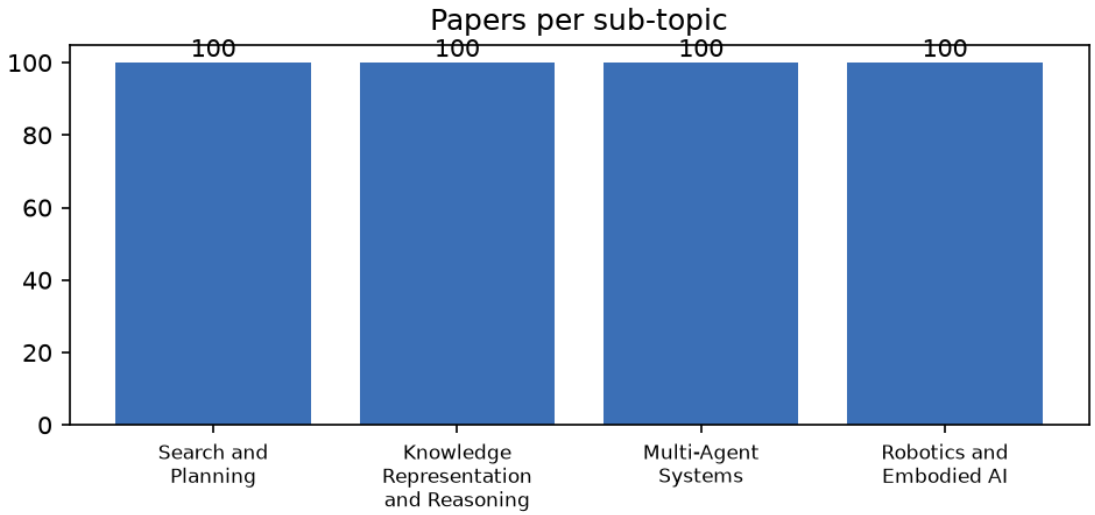
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1. Overview

Metric	Value
Total papers (full text available)	400
arXiv category	cs.AI (papers listing cs.AI: 400 of 400)
Sub-topics	4
Papers matching more than one sub-topic	0
Publication date range	2021-02-05 to 2026-10-15
Total pages	7987
Pages per paper (mean / median / min / max)	19.97 / 16.0 / 2 / 346
Total words	4,251,096
Words per paper (mean / median)	10,628 / 8,855
Unique author names	1831

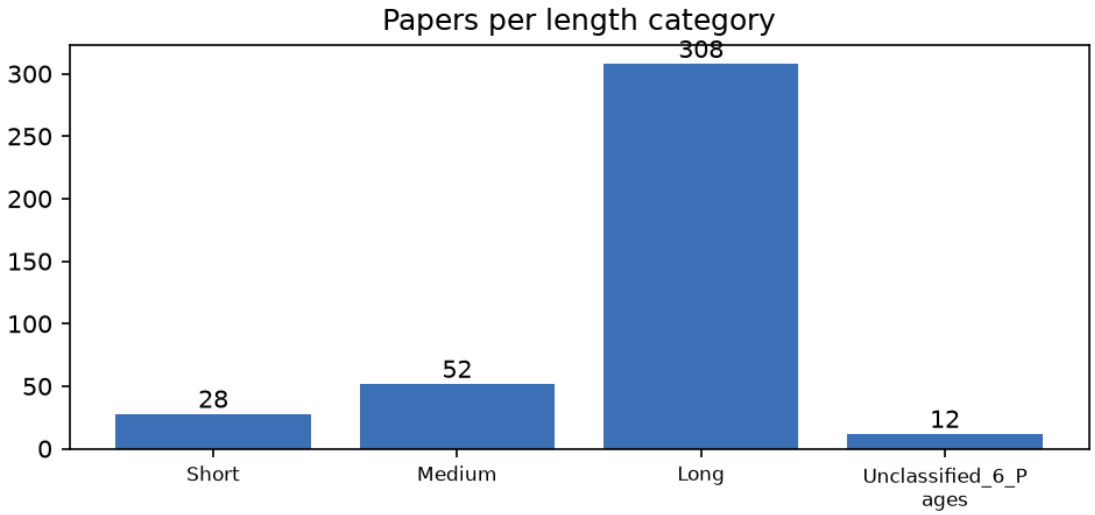
2. Distribution by sub-topic

Sub-topic	Papers	Share
Search and Planning	100	25.0%
Knowledge Representation and Reasoning	100	25.0%
Multi-Agent Systems	100	25.0%
Robotics and Embodied AI	100	25.0%



3. Document length categories

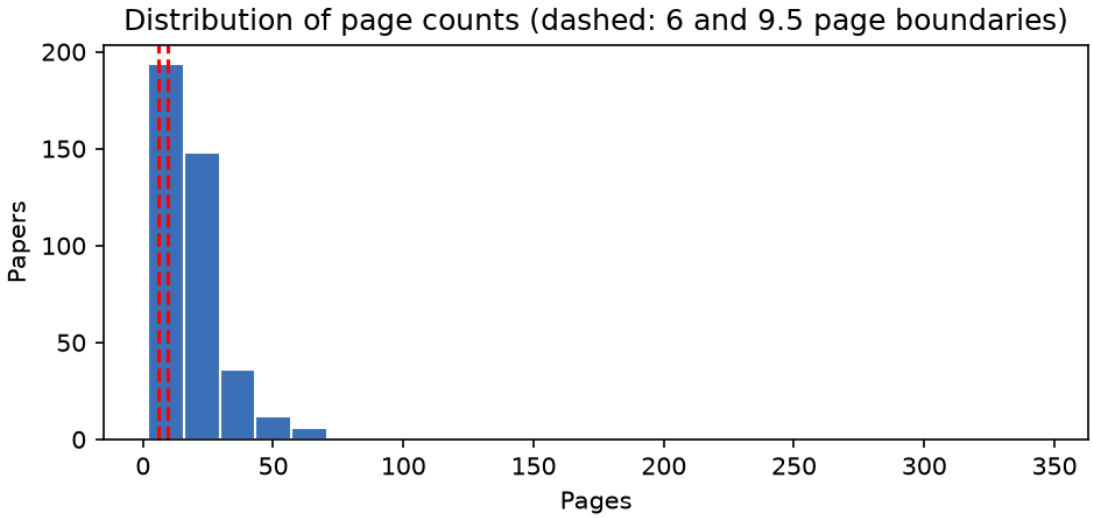
Category	Definition	Papers	Share
Short	Fewer than 6 pages	28	7.0%
Medium	7 to 9 pages	52	13.0%
Long	More than 9 pages	308	77.0%
Unclassified_6_Pages	Exactly 6 pages (not defined in the assignment; kept, pending instructor)	12	3.0%



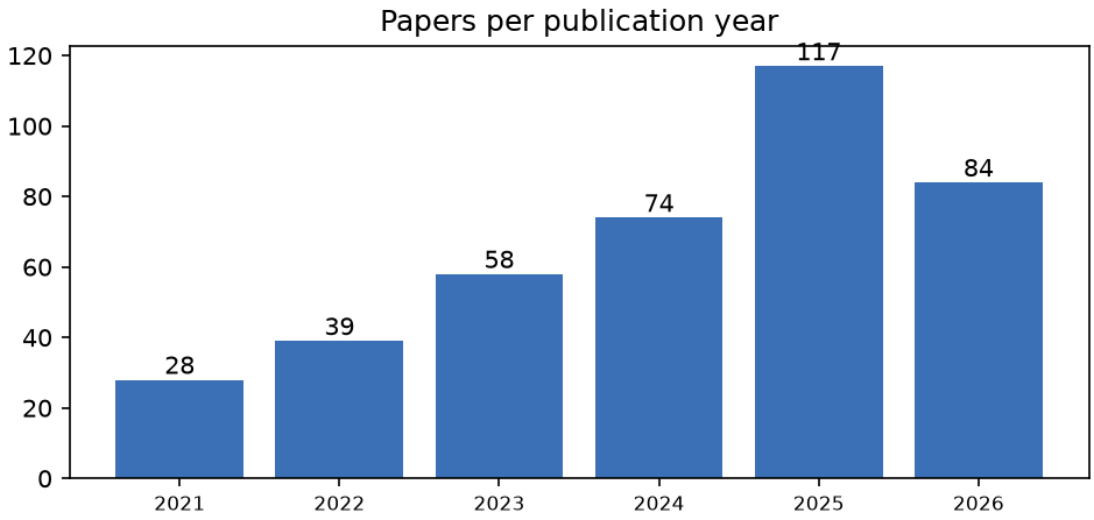
Length category by sub-topic

Sub-topic	Short	Medium	Long	Unclassified_6_Pages	Total
Search and Planning	8	20	67	5	100
Knowledge Representation and Reasoning	6	12	81	1	100
Multi-Agent Systems	3	10	85	2	100
Robotics and Embodied AI	11	10	75	4	100

4. Pages and text length



5. Publication year



6. Collection details and reproducibility

Field	Value
category	cs.AI
date_range	{"from": "2021-01-01", "to": "2026-10-31"}
collection_timestamp_utc	2026-11-01T21:28:03.282343+00:00
target_per_topic	100
total_unique_papers	404
topic_counts	{"Search and Planning": 100, "Knowledge Representation and Reasoning": 104, "Multi-Agent Systems": 100, "Robotics and Embodied AI": 100}
queries	{"Search and Planning": "cat:cs.AI AND (all:\"automated planning\" OR all:\"heuristic search\" OR all:\"search-based planning\") AND submittedDate:[202101010000 TO 202610312359]", "Knowledge Representation and Reasoning": "cat:cs.AI AND (all:\"knowledge representation\" OR all:\"automated reasoning\" OR all:\"knowledge reasoning\") AND submittedDate:[202101010000 TO 202610312359]", "Multi-Agent Systems": "cat:cs.AI AND (all:\"multi-agent systems\" OR all:\"multi agent\" OR all:\"multiagent systems\") AND submittedDate:[202101010000 TO 202610312359]", "Robotics and Embodied AI": "cat:cs.AI AND (all:\"embodied AI\" OR all:\"robot learning\" OR all:\"robotic planning\") AND submittedDate:[202101010000 TO 202610312359]"}
source	arXiv API
metadata_complete	True
successful_extractions	400
collection_updated_utc	2026-11-01T22:24:19.264287+00:00

7. Data quality notes

- Four papers from the first collection failed to download (HTTP 404) and were replaced by four Knowledge Representation and Reasoning papers; the failed records are kept in the raw logs for documentation only and are excluded from the final corpus.

- 12 papers have exactly 6 pages. The assignment does not state which length category they belong to, so they are kept under a separate label pending instructor clarification.

- All papers in the final corpus have extracted full text; none are empty.